

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Update, Change]	257375	53.227681
1	[Git, Change, Create]	74452	15.397406
2	[Git, Change, Delete]	44503	9.203658
3	[Git, Change, Rename]	23284	4.815360
4	[File, Git]	15911	3.290551
5	[Git, Commit]	15582	3.222511
6	[Java, ByteCode, Parameter]	9227	1.908234
7	[Java, ByteCode, Member, Method]	8877	1.835851
8	[Java, ByteCode, Bound]	6300	1.302902
9	[Java, Value, ByteCode, Annotation]	5334	1.103124
10	[Java, ByteCode, ParameterizedType, Bound]	4849	1.002821
11	[Xml, Element]	2384	0.493035
12	[Java, ByteCode, Member, Field]	2180	0.450845
13	[Xml, Text]	1596	0.330068
14	[Java, ByteCode, Member, Constructor, Method]	1577	0.326139
15	[Java, ByteCode, Bound, WildcardType]	968	0.200192
16	[Java, ByteCode, Member, Method, Lambda]	851	0.175995
17	[Java, ByteCode, Bound, TypeVariable]	830	0.171652
18	[Type, File, Java, ByteCode, JavaType]	689	0.142492
19	[Type, File, Java, Class, ByteCode]	647	0.133806
20	[Json, Key]	533	0.110230
21	[Value, Json, Scalar]	523	0.108162
22	[Git, Change, Copy]	470	0.097201
23	[Type, File, Java, ByteCode, ResolvedDuplicate...]	459	0.094926
24	[Java, ByteCode, Member, Method, GenericDeclar...]	455	0.094098
25	[Java, Value, ByteCode, Primitive]	322	0.066593
26	[Author, Git, Person]	302	0.062457
27	[Type, File, Java, ByteCode, ExternalType]	301	0.062250
28	[Committer, Git, Person]	246	0.050875
29	[Maven, Dependency]	192	0.039707

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

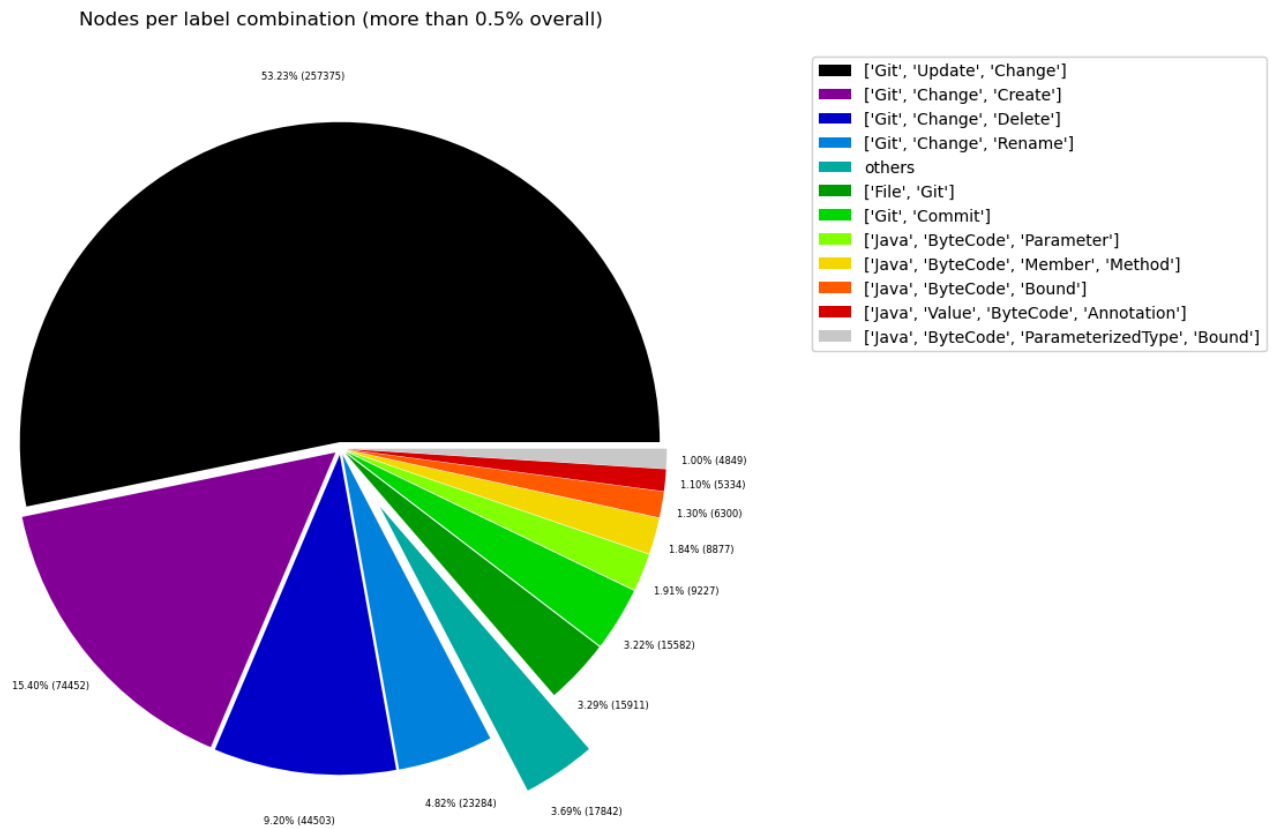


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, jQAssistant]	1	0.000207
1	[Git, Branch]	1	0.000207
2	[Repository, File, Git]	1	0.000207
3	[File, Json]	2	0.000414
4	[File]	3	0.000620
5	[Type, File, Java, ByteCode, Record, GenericDe...	4	0.000827
6	[Maven, Exclusion]	4	0.000827
7	[Value, Array, Json]	8	0.001654
8	[Type, File, Java, ByteCode, Throwable, Extern...	8	0.001654
9	[Java, ByteCode, Member, Constructor, Method, ...	9	0.001861
10	[Type, File, Java, ByteCode, Throwable, Resolv...	11	0.002275
11	[Maven, Scm]	11	0.002275
12	[File, Maven, Xml, Pom, Document]	11	0.002275
13	[Type, File, Java, ByteCode, Void]	11	0.002275
14	[Java, ManifestSection]	11	0.002275
15	[File, Java, Manifest]	11	0.002275
16	[File, Artifact, Jar, Archive, Zip, Java]	11	0.002275
17	[File, Java, ServiceLoader]	12	0.002482
18	[File, Java, Properties]	13	0.002689
19	[Maven, ExecutionGoal]	13	0.002689
20	[Maven, PluginExecution]	13	0.002689
21	[Type, File, Java, ByteCode, Enum]	17	0.003516
22	[jQAssistant, Rule, Concept]	19	0.003929
23	[Maven, Configuration]	20	0.004136
24	[Maven, Plugin]	20	0.004136
25	[Xml, Attribute]	22	0.004550
26	[Type, File, Java, ByteCode, PrimitiveType]	39	0.008066
27	[Type, File, Java, Class, ByteCode, Throwable]	42	0.008686
28	[Type, File, Java, ByteCode, Annotation]	42	0.008686
29	[Xml, Namespace]	44	0.009100

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

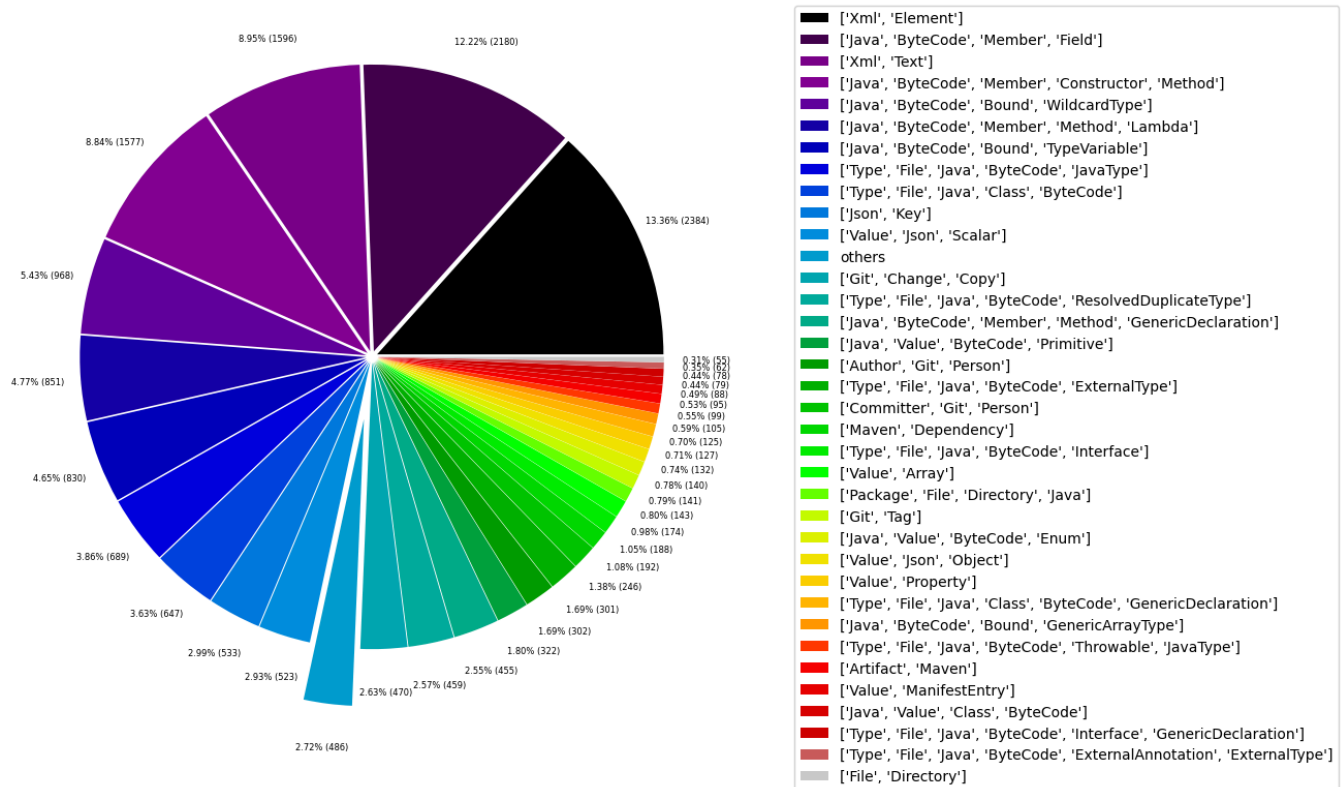


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

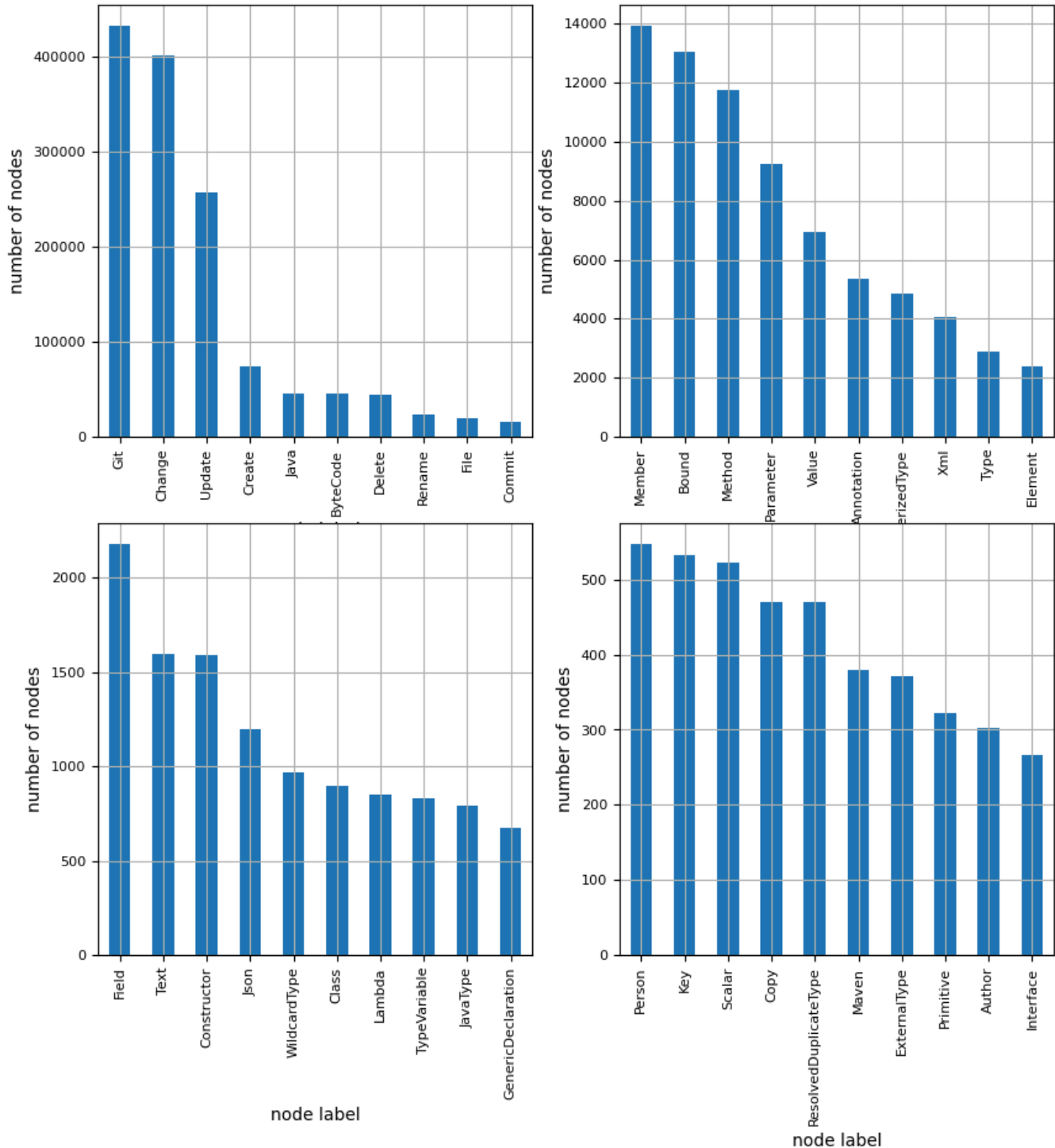
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	432268	89.397273
1	Change	400084	82.741306
2	Update	257375	53.227681
3	Create	74452	15.397406
4	Java	45178	9.343255
5	ByteCode	44977	9.301686
6	Delete	44503	9.203658
7	Rename	23284	4.815360
8	File	19047	3.939107
9	Commit	15582	3.222511
10	Member	13949	2.884790
11	Bound	13052	2.699282
12	Method	11769	2.433945
13	Parameter	9227	1.908234
14	Value	6927	1.432572
15	Annotation	5376	1.111810
16	ParameterizedType	4849	1.002821
17	Xml	4057	0.839027
18	Type	2874	0.594371
19	Element	2384	0.493035
20	Field	2180	0.450845
21	Text	1596	0.330068
22	Constructor	1586	0.328000
23	Json	1198	0.247758
24	WildcardType	968	0.200192
25	Class	893	0.184681
26	Lambda	851	0.175995
27	TypeVariable	830	0.171652
28	JavaType	788	0.162966
29	GenericDeclaration	671	0.138769
30	Person	548	0.113332
31	Key	533	0.110230
32	Scalar	523	0.108162
33	Copy	470	0.097201
34	ResolvedDuplicateType	470	0.097201
35	Maven	379	0.078381
36	ExternalType	371	0.076726
37	Primitive	322	0.066593
38	Author	302	0.062457
39	Interface	266	0.055011

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 1501814

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	400084	26.640050
1	MODIFIES	400084	26.640050
2	UPDATES	257375	17.137608
3	CREATES	98206	6.539159
4	DELETES	67787	4.513675
5	COMMITTED	31164	2.075091
6	RENAMES	23284	1.550392
7	INVOKES	22194	1.477813
8	HAS_PARENT	18926	1.260209
9	OF_TYPE	18495	1.231511
10	DEPENDS_ON	17561	1.169319
11	HAS_FILE	15911	1.059452
12	HAS_COMMIT	15582	1.037545
13	DECLARES	14288	0.951383
14	OF_RAW_TYPE	13122	0.873743
15	HAS	9843	0.655407
16	HAS_NEW_NAME	9378	0.624445
17	RETURNS	8100	0.539348
18	READS	5919	0.394123
19	HAS_ACTUAL_TYPE_ARGUMENT	5753	0.383070
20	ANNOTATED_BY	5333	0.355104
21	OF_GENERIC_TYPE	4392	0.292446
22	CHANGED_TOGETHER_WITH	3686	0.245437
23	SIMILAR	3370	0.224395
24	CONTAINS	3183	0.211944
25	RETURNS_GENERIC	2586	0.172192
26	RESOLVES_TO	2566	0.170860
27	HAS_FIRST_CHILD	2384	0.158741
28	HAS_LAST_CHILD	2384	0.158741
29	HAS_ELEMENT	2362	0.157276

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

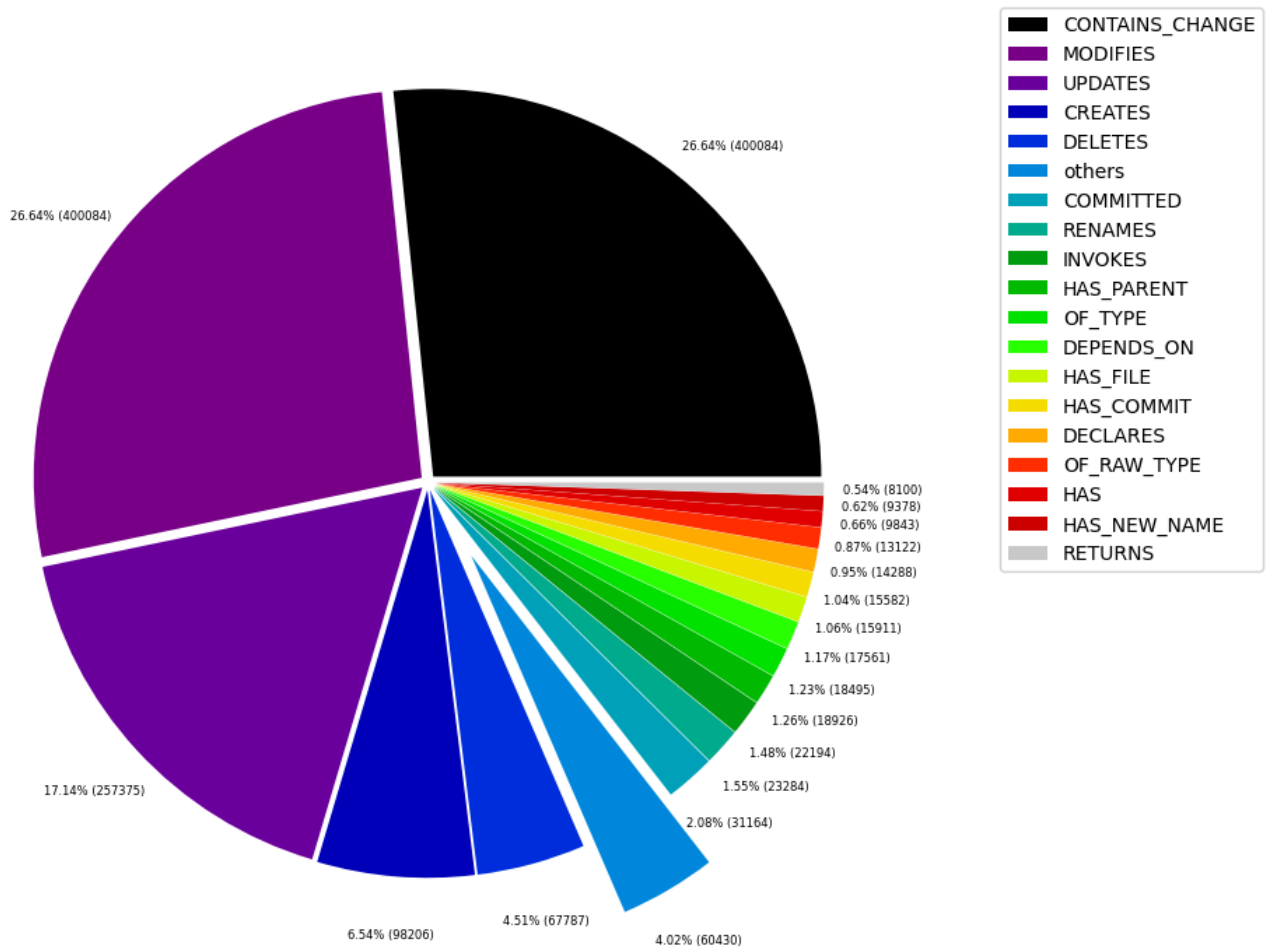


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

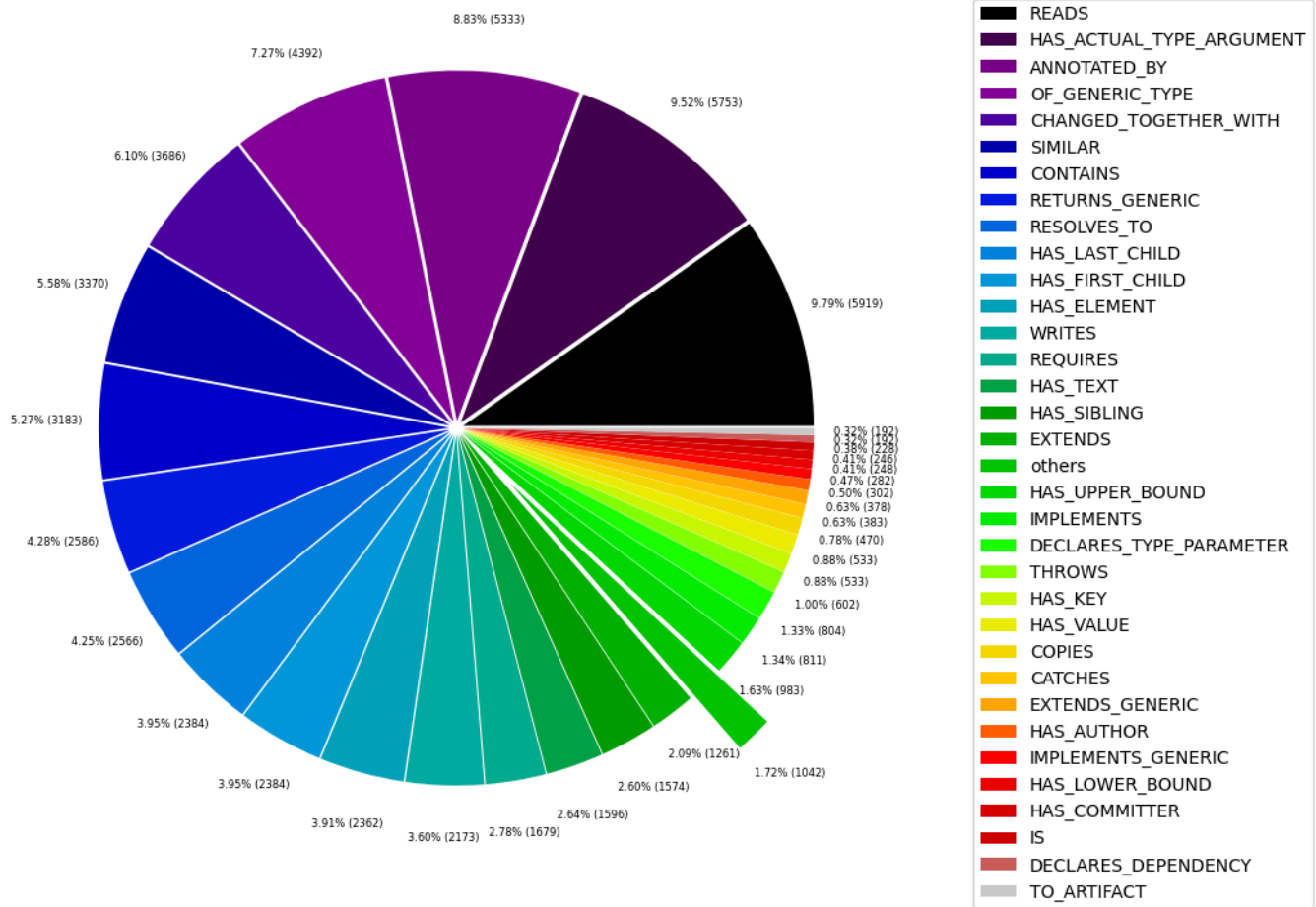
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_PROPERTY	1	0.000067
1	HAS_BRANCH	1	0.000067
2	HAS_HEAD	2	0.000133
3	THROWS_GENERIC	3	0.000200
4	EXCLUDES	4	0.000266
5	HAS_SCM	11	0.000732
6	DESCRIBES	11	0.000732
7	HAS_ROOT_ELEMENT	13	0.000866
8	HAS_GOAL	13	0.000866
9	HAS_EXECUTION	13	0.000866
10	INCLUDES_CONCEPT	19	0.001265
11	IS_ARTIFACT	20	0.001332
12	HAS_CONFIGURATION	20	0.001332
13	USES_PLUGIN	20	0.001332
14	OF_NAMESPACE	22	0.001465
15	HAS_ATTRIBUTE	22	0.001465
16	REQUIRES_TYPE_PARAMETER	26	0.001731
17	REQUIRES_CONCEPT	28	0.001864
18	DECLARES_NAMESPACE	44	0.002930
19	HAS_DEFAULT	53	0.003529
20	HAS_COMPONENT_TYPE	105	0.006992
21	CONTAINS_VALUE	128	0.008523
22	HAS_TAG	141	0.009389
23	ON_COMMIT	141	0.009389
24	COPY_OF	181	0.012052
25	TO_ARTIFACT	192	0.012785
26	DECLARES_DEPENDENCY	192	0.012785
27	IS	228	0.015182
28	HAS_COMMITTER	246	0.016380
29	HAS_LOWER_BOUND	248	0.016513

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numberO
0	[Git, Commit]	CONTAINS_CHANGE	[Git, Update, Change]	257375	15582	
1	[Git, Update, Change]	MODIFIES	[File, Git]	257375	257375	
2	[Git, Update, Change]	UPDATES	[File, Git]	257375	257375	
3	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	74452	15582	
4	[Git, Change, Create]	MODIFIES	[File, Git]	74452	74452	
5	[Git, Change, Create]	CREATES	[File, Git]	74452	74452	
6	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	44503	15582	
7	[Git, Change, Delete]	MODIFIES	[File, Git]	44503	44503	
8	[Git, Change, Delete]	DELETES	[File, Git]	44503	44503	
9	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	23284	15582	
10	[Git, Change, Rename]	CREATES	[File, Git]	23284	23284	
11	[Git, Change, Rename]	RENAMES	[File, Git]	23284	23284	
12	[Git, Change, Rename]	DELETES	[File, Git]	23284	23284	
13	[Git, Change, Rename]	MODIFIES	[File, Git]	23284	23284	
14	[Git, Commit]	HAS_PARENT	[Git, Commit]	18915	15582	
15	[Repository, File, Git]	HAS_FILE	[File, Git]	15911	1	
16	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	15582	1	
17	[Author, Git, Person]	COMMITTED	[Git, Commit]	15582	302	
18	[Committer, Git, Person]	COMMITTED	[Git, Commit]	15582	246	
19	[Java, ByteCode, Member, Method]	INVOKES	[Java, ByteCode, Member, Method]	13279	8849	
20	[File, Git]	HAS_NEW_NAME	[File, Git]	9378	15911	
21	[Java, ByteCode, Member, Method]	HAS	[Java, ByteCode, Parameter]	5325	8849	
22	[Java, ByteCode, Member, Method]	READS	[Java, ByteCode, Member, Field]	5320	8849	
23	[Java, Value, ByteCode, Annotation]	OF_TYPE	[Type, File, Java, ByteCode, ExternalAnnotatio...	4990	5334	
24	[Java, ByteCode, Parameter]	OF_TYPE	[Type, File, Java, ByteCode, javaType]	4105	9227	
25	[Java, ByteCode, Parameter]	ANNOTATED_BY	[Java, Value, ByteCode, Annotation]	3797	9227	
26	[File, Git]	CHANGED_TOGETHER_WITH	[File, Git]	3068	15911	
27	[Java, ByteCode, ParameterizedType, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, javaType]	2799	4849	
28	[Xml, Element]	HAS_ELEMENT	[Xml, Element]	2362	2384	
29	[Java, ByteCode, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, javaType]	2256	6300	

Graph Density

```
total_number_of_nodes (vertices): 483536
total_number_of_relationships (edges): 1501814
-> total directed graph density: 6.423317947431575e-06
-> total directed graph density in percent: 0.0006423317947431575
```